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Neural cell state shifts and fate loss in ageing and age-related diseases

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Abstract

Most age-related neurodegenerative diseases remain incurable owing to an incomplete understanding of the disease mechanisms. Several environmental and genetic factors contribute to disease onset, with human biological ageing being the primary risk factor. In response to acute cellular damage and external stimuli, somatic cells undergo state shifts characterized by temporal changes in their structure and function that increase their resilience, repair cellular damage, and lead to their mobilization to counteract the pathology. This basic cell biological principle also applies to human brain cells, including mature neurons that upregulate developmental features such as cell cycle markers or glycolytic reprogramming in response to stress. Although such temporary state shifts are required to sustain the function and resilience of the young human brain, excessive state shifts in the aged brain might result in terminal fate loss of neurons and glia, characterized by a permanent change in cell identity. Here, we offer a new perspective on the roles of cell states in sustaining health and counteracting disease, and we examine how cellular ageing might set the stage for pathological fate loss and neurodegeneration. A better understanding of neuronal state and fate shifts might provide the means for a controlled manipulation of cell fate to promote brain resilience and repair.

Introduction

Theories of age-related diseases are often based on the ‘multiple hit’ hypothesis, where one hit comes from cell-intrinsic or environmental factors and an additional, essential hit comes from progressive ageing[1–4]. Therefore, ageing is one of the largest risk factors for neurodegeneration, many cancers, and other diseases of immense societal impact[4–6]. Even in the case of inherited, early-onset Alzheimer disease (AD), amyotrophic lateral sclerosis and Parkinson disease, neurons in the brain function and survive for decades before disease onset. The healthy young brain is frequently exposed to stress and damage (for example,

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high levels of reactive oxygen species (ROS) and calcium) that lead to macromolecular and organellar damage during periods of high activity; however, it possesses resilience, the capacity for repair, and adequate mechanisms to deal with such disease triggers[7,8]. In addition, as a response to acute cellular damage and external stimuli, glial cells and postmitotic neurons frequently undergo state shifts (that is, large, abrupt, persistent changes in the structure and function of a cell) to increase their resilience, to adapt their function to environmental changes, to repair themselves, or to counteract pathology[9–13]. For instance, in response to injury, the mature cell fate of neurons can be suppressed and they enter a regenerative state[9,14]. Such temporary shifts overall sustain function and resilience in the young human brain, and the cells return to their normal somatic functions afterwards.

In the old brain, an age-related loss of neuronal resilience and molecular and epigenetic entropy provide a cellular platform for disease triggers to escalate[15,16]. During ageing, the cell regulatory epigenetic machinery becomes less efficient and is unable to correct genomic alterations, leading to an altered epigenetic status characterized by the loss of regulatory epigenetic landmarks[17]. In fact, cells become less able to properly respond or adapt to intrinsic and extrinsic signals, which ultimately results in loss of phenotypic plasticity[18]. Stress-induced state shifts that permit repair of a young cell are thought to culminate in irreversible loss of cellular identity in an old cell with an eroded epigenetic landscape and, eventually, lead to chronic brain dysfunction and terminal neurodegeneration. Human biological ageing is the primary risk factor for the development of neurodegenerative diseases, as it is for the development of many cancers[19]. However, there have been few insights into the molecular and cellular convergence of disease pathways involved in normal ageing, neurodegeneration and cancer.

In this Perspective, we collect and discuss the evidence that identifies loss of cellular identity as a fundamental underlying feature of age-related neurodegenerative disorders. To do so, we first define and distinguish cell states from cell fates (see Box 1), and provide examples of physiological state shifts of neuronal and glial cells in response to disease-related stimuli. We next discuss how ageing-related epigenetic erosion predisposes old cells to fate shifts through loss of identity-defining epigenetic and molecular signatures, and how old cells are more likely to withdraw from their somatic identity than young cells experiencing the same stressors. Thus, chronic state shifts might escalate into a terminal fate loss of neurons, which ultimately displays as neurodegeneration; we further note that similar molecular mechanisms promote carcinogenesis in many other somatic cell types. Finally, we summarize emerging biotechnological concepts of how controlled state shifts via partial reprogramming can be applied to hijack cellular pathways to induce young-like resilience and repair, and we explore how such concepts could be tailored to neurons.

Cell states and fates in the brain

During organismal development, stem cells differentiate and commit to specific cell fates. Cell fate describes developmental identity of a cell population, which does not allow dynamic shifting between different fates. A cell fate is, for example, a neuron, an astrocyte or a microglial cell. Such cell fates were once typically considered to be hard-wired, but single cell technology has now revealed a high level of transcriptional and functional

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variation among cells within a specific fate, calling into question the established ideas of cell fate[20] (see Box 1). Defined cell fates often coincide with histologically defined tissues and developmental lineages and are also called ‘cell type’ or ‘identity’. Cells of the same fate have many common characteristics (for example, epigenomic, transcriptomic and metabolomic signatures) combined with morphology and/or function. Although cells with the same identity share these common features, they remain highly heterogeneous; therefore we refer to this as a ‘cloud’ of identity[21–23] (Fig. 1). During neural development, neural stem cells are anchored in an identity cloud defined by multiple layers of concerted signatures on epigenomic, transcriptomic, proteomic and metabolomic levels. In response to developmental cues, neural stem cells eventually abandon their cloud, progressively changing their fate until they commit to neuronal or glial cell fates[20] (Fig. 1). Each cell fate has specific metabolic preferences that are not static and can switch in response to external or internal stimuli.

Metabolism is a powerful instructor of cell fate decisions during development and helps to maintain cellular integrity and resilience through adaptations to nutrient and oxygen availability and cellular energetic needs. Metabolism also helps to restructure gene expression profiles[24], which results in a high cell-to-cell heterogeneity within a specific cell fate identity cloud; this heterogeneity represents the various cell states[20,21]. Similar to fates, cellular states are defined by concerted multilayered molecular, morphological and functional changes of a cell, but always within the boundaries that define a certain cell fate[22] (see Box 1).

A shift of a cell from one state to another can be described as a temporary molecular shift on various levels in a dynamic system that is reversible. These shifts might occur on a transcriptional, metabolic or morphological level to put the cell in a position to temporarily fulfil a specific function. Some better-studied examples of physiological state changes in response to stimuli are changes between a quiescent and proliferative stem cell[25], glucose activation of insulin-producing pancreatic β-cells[26], changes in noradrenergic neurons between sleep and wakefulness[27], and immune cell activation[28]. Advances in single cell transcriptomes have provided an updated view in which cells exist in the continuum of a dynamic landscape; thus, cell fates can be viewed as a collection of cell states[22,29]. The discussion about the differences between state and fate has thus become more quantitative than qualitative. In that regard, fate changes describe the process in which a collection of state changes occur, and the cell eventually departs from its identity cloud. However, the high resolution provided by single cell transcriptomics, and the dynamic cellular transcriptional responses have also encouraged the assumption of cell fate heterogeneity, where cells can dynamically shuttle between mild state changes.

Cellular state shifts in the brain

Activation of microglia and astrocytes

State shifts have an important role in how cells and organs respond to stress and injury, aiding repair and resilience. One of the best studied state shifts in the brain is the ‘activation’ of microglia and astrocytes[30,31]. Historically, three different microglial states and two different astrocytic states have been distinguished. First, resting microglia or neuroprotective

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astrocytes are present under physiological conditions, providing neurotrophic factors, supporting neurotransmitter recycling and aiding synapse formation. Second, in response to injury, pathogens or pathologies, microglia and astrocytes become activated and enter a pro-inflammatory state. Pro-inflammatory glial cells secrete chemokines and immune-regulatory molecules that can result in the removal of damaged neurons. Healthy mature neurons are thought to be resistant to such fatal cues, as they develop profound apoptotic brakes[8,32–34]. Thus, neurotoxic glial cells might only remove harmful neurons, such as senescent neurons secreting cytotoxic factors. Furthermore, reactive and not necessarily pro-inflammatory microglia also shape neuronal circuitry by eliminating (or 'pruning') inferior synapses during development and adult brain function[35,36]. Last, anti-inflammatory microglia, also called phagocytic microglia, secrete neuroprotective molecules and phagocytose cell debris[37].

With the advances in single cell technologies, more cellular states of glial cells have been discovered, with specific transcriptional profiles depending on their function. No single state of astrocytes is neuroprotective or inflammatory; instead, several states exist for each function, which together form a dynamic system[38]. Although the emergence of specific glial states in response to acute injury or pathogens is temporary, chronic activation results in reactive disease states and tissue dysfunction[10]. Damage-associated microglia and reactive astrocytes have been observed in the brain of individuals with neurodegenerative disorders[39–42]. This accumulation of glial cells in an inflammatory state in neurodegenerative disorders suggests that pathological state changes can be irreversible. Thus, pathological state changes can equally be considered fate changes, as they are not temporary.

Cellular senescence

One of the most famous examples of a state change that shares features with a fate change is the transition to cellular senescence. Senescence was originally described in 1961 as an irreversible proliferative arrest in human skin cells following serial passage, a finding that changed the conventional wisdom at the time that proliferating cells could divide indefinitely in vitro[43]. In this case, stress to the cell would be in the form of telomere attrition, where progressive erosion of the ends of chromosomes eventually triggers a DNA damage response that induces the expression of proteins such as P53, P16 and P21, and permanently forces the cell into a new, proliferatively arrested state of senescence[44,45]. This transition has been particularly noteworthy in the ageing field, because the capacity to transition to senescence seems to be preserved throughout most mammalian cell types both in vivo and in vitro and the changes that are initiated following senescence can be quite dramatic[46,47]. Following senescence, cells experience wide-ranging changes, including modifications of cell morphology and chromatin organization, apoptosis resistance, lysosomal modifications and an inflammatory phenotype[44]. In contrast to the exit of the cell cycle in quiescent stem cells, senescence is thought to be irreversible, putting it in between state and fate changes[48]. Although many forms of senescence are adaptive early in life, during development or as a tumour suppressor, a persistence of cells transitioning into senescence is associated with tissue dysfunction[49,50]. Indeed, although cellular replication and telomere dysfunction are major contributors to senescence, studies have demonstrated that a wide range of stressors

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or signals can trigger senescence[51]. Given that these triggers include oncogene activation and a shift in metabolic state[52,53], which many cell types experience during ageing, it has become more widely accepted that even non-proliferating, postmitotic cells can enter senescence[54,55]. Indeed, senescence in neurons has been observed in many contexts associated with ageing, particularly in neurodegeneration[56–59] and injury[60,61].

Postmitotic neurons have evolved to survive for decades[32]. However, they have high energy demands that result in an accumulation of oxidized products with age. Rather than undergoing cell death, postmitotic neurons might enter senescence to maintain neuronal network integrity, although this hypothesis remains to be tested. Neuronal senescence, like senescence in other cell types, has no single universally reliable marker, possibly owing to the large number of neuronal subtypes and the predominant reporting of senescence markers in dividing cell types. Nevertheless, many markers of senescence have been observed in neurons including increased DNA damage, high levels of ROS, oxidative damage, genetic markers such as expression of *P53, P16, P21* and inflammatory genes, and elevated levels of senescence-associated β-galactose[55,56,59,62,63]. As senescence-triggering stressors increase in frequency during ageing, so too do the proportions of senescent-like neurons in the brain of aged animals[56,62,64]. Here, a state transition to senescence in neurons seems to be a response to damage, an attempt to increase resilience, and a starting point for inflammatory cascades. Neuronal senescence thus highlights how the state transition of even a small number of cells in the brain can have a profound and lasting impact on tissue homeostasis, highlighting the need to investigate the molecular basis and outcomes of state transitions in neural cells.

Fate loss in ageing and disease

All organisms age, but they all do it at different rates. Increased age is associated with higher mortality rates and thus interest in understanding the mechanisms of ageing, with the aim of slowing the ageing process, has grown. Beginning in the 1930s, researchers have been able to extend the lifespan of model organisms through dietary restriction[65–69]. With the arrival of advanced analysis tools and more appropriate model systems in the 1990s and early 2000s, interest shifted towards the molecular changes of ageing, resulting in the discovery of genetic variants and chemical compounds that promote longevity in model organisms[70–73]. From there, the concept of fate instability has evolved from age-related epigenetic drift[74–76], to transcriptional noise[77], and further to the current concepts that describe how fate is a fundamental feature of ageing instability in yeast[78] and mammals (preprint)[16]. A current model of ageing describes the gradual erosion of the epigenetic landscape, or epigenetic entropy, that results in fate instability and fate loss over time[16,74]. These hypotheses from the ageing field can be further anchored in well-established literature from other fields. The clearest parallels can be drawn with cancer. First, the incidence of many cancers is age-related[79]. Second, the markers of fate loss and the mechanisms by which fate loss is orchestrated during cancer transformation are similar to those of fate loss in age-related neurodegeneration. Cancer cells regain features of development and, to a degree, mirror progenitor states during development[80,81], including open chromatin around transcription factors involved in stemness or proliferation[82], or a metabolism focusing on the generation of macromolecules and intermediates for epigenetic control[83,84]. However, although during

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development the progenitor state is temporary and follows a tightly orchestrated timing – for example, by the use of bivalent promoter chromatin marks[85,86] – the shift towards a progenitor-like state in cancer cells is a stable and irreversible fate change[82,87].

In fact, when approached from this perspective, physiological state shifts and pathological fate instability in the brain have been described for decades (Box 1). For example, evidence indicates that transient phosphorylation of tau and the formation of non-toxic aggregates are required for axonal outgrowth and synapse formation in the developing brain, whereas sustained hyperphosphorylation of tau is a hallmark of AD and contributes to fate loss[88–90]. Additionally, in human post-mortem studies, expression of cell cycle markers was higher in neurons from individuals with AD than in neurons from cognitively healthy age-matched controls[91]. This observation led to the cell cycle re-entry hypothesis[91], which suggests that mature, postmitotic neurons initiate transcriptional programmes of proliferaion usually only seen in precursor cells. Interestingly, these signs of cell cycle re-entry of neurons in AD have not been interpreted in the context of cancer-like de-differentiation and fate loss. However, this interpretation should be re-examined. Indeed, such changes could constitute a state shift continuum that has the potential to culminate in the induction of senescence.

In conclusion, the emerging concept of cell fate instability during ageing provides a new perspective that invites a re-visit of previous observations and hypotheses. Thanks to advances over the past 10 years, ageing research is now an established and quantitative field at the molecular and cellular levels. Research into human ageing has now made outstanding advances towards understanding how resilience is compromised in ageing. However, the functional relationship between human biological ageing and the onset of age-related diseases is still obscure. The advances of model systems to mimic human ageing have started to transform our perspective on the interface between ageing and neurodegeneration, and they hint at a cell state and identity crisis that undermines neuronal resilience[32,92,93].

Damage-induced de-differentiation

The cell biological principle that damage causes de-differentiation is a well-studied and conserved principle from plants[94] to amphibians[95]; it has even been described in regenerative organs, such as the liver, in humans[96,97]. Damage-induced de-differentiation followed by cellular rejuvenation and re-differentiation (DiD-RR) is the endogenous mechanism by which cells regenerate in response to acute injuries and ageing damage[98–102]. DiD-RR is a highly conserved mechanism by which somatic cells temporarily regress towards an immature-like state of resilience and rejuvenation and then eventually re-differentiate to again undertake their somatic function (Fig. 2). Organ regeneration in amphibians is based on a temporary de-differentiation of fully differentiated somatic cells to multipotent stem cells. In addition to re-entering the cell cycle[103], they initiate a precisely timed gene expression programme to upregulate stemness-associated genes, such as *Myc, Klf4 and Sox2* (ref. 104), and induce a metabolic shift to aerobic glycolysis, which is required to make fully differentiated cells amenable to de-differentiation[24]. Despite the dynamic transcriptional and metabolic changes occurring during this regeneration to resemble early progenitor cells, many histone modifications are kept stable, forming an epigenetic memory – the anchor in their identity cloud[105]. Thus, generated multipotent cells

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are only able to give rise to specific cell types of the regenerating organ, regressing to their original identity cloud after full regeneration[106].

The regenerative capacity in mammals is known to decrease with age, caused in part by dysfunction of the stem cell pool required for regeneration of specific tissues, such as muscle[107,108]. However, evidence suggests that the process of DiD-RR also becomes less efficient or blocked as animals age, resulting, for example, in impaired regeneration of the distal digit tip in old compared with young mice[109]. Aged cells exhibit epigenetic and metabolic drift over time, as bivalent promoter chromatin patterns become reinforced and irreversible[87] and accumulated stress results in an energy crisis[109], rendering cells epigenetically and metabolically inflexible. However, it remains to be ascertained how changes observed during ageing and disease compromise the process of DiD-RR, making it less efficient overall or leaving cells stuck in intermediate phases.

Compared with most other somatic cells, neurons in the adult brain are special in that their functional importance for the organism is largely owing to their connectivity. If a neuron were to follow an amphibian-like DiD-RR process, it might produce two or more new neurons, but these would have irreversibly lost the connectivity and information of the mother cell[110]. Thus, human brain neurons were long thought to lack DiD-RR and the ability to regress into a bona fide cell cycle. Surprisingly, findings in induced pluripotent stem cells (iPSCs) and induced neuronal models of AD (see Box 2 for definition) published in the last few years indicate that mature human neurons react to acute and chronic injury by similar mechanisms, which have been described as partial de-differentiation[92,111]. Characteristics of these AD models, such as glycolytic reprogramming[93,112], cell cycle re-entry[110,113] or increased levels of hyperphosphorylated tau[90,114,115], mark a de-differentiated neuronal state that might have protective effects in response to stress. Studies of peripheral nerve and spinal cord injury in mice have also revealed a regenerative transcriptional state of damaged neurons, characterized by neuronal fate loss and the re-activation of stem cell-like gene programmes[9,14,116]. These findings suggest that fate loss is important for tissue maintenance under stress, where a neuron can lose either its connectivity or its resilience.

Although the existence of complete DiD-RR in postmitotic neurons is a matter of ongoing debate, chronic de-differentiation and fate loss are evident in neurodegenerative diseases[92,117,118]. For example, induced neurons from individuals with AD show elevated levels of ROS and DNA damage markers[92]. In response to these stressors, they gain a de-differentiated transcriptional and morphological phenotype, expressing signalling pathways characteristic of stem cells and cancers[92,111]. AD neurons further alter their bioenergetic glucose signature by shifting to increased aerobic glycolysis while maintaining mitochondrial oxidation[93]. This metabolic shift (similar to the Warburg effect) restructures the epigenetic landscape so that the cell acquires a progenitor-like phenotype. As such, AD neurons largely show an epigenetic ageing signature that reflects a de-differentiated cellular state and promotes transcriptional programmes that ultimately lead to cell identity loss. These large epigenetic changes might be described as epigenetic entropy. They allow the binding of regulatory factors to genomic regions that in healthy cells are not accessible, including metabolic regulators *LDHA* and *HIF1A*, and cancer-associated regulators such

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as *YY1* or *STAT1* (ref. 92). The expression of these genes, which regulate non-neuronal functions, apparently primes the cells to enter a less mature, de-differentiated cellular state. The de-differentiation observed in AD neurons might initially be activated by the cells as a temporary state. However, the age-related epigenetic drift, metabolic crisis and consequential entropy are likely to result in an increased number of 'uncontrolled' side effects and ultimately lead to irreversible fate loss.

Even though the cancer and AD phenotypes seem to be quite different, the epigenetic and molecular events occurring in cancer cells and in AD neurons share common molecular mechanisms that result in fate loss, a finding that might open new routes for therapeutic intervention. Although the activation of stemness-associated and proliferation genes results in uncontrolled proliferation and malignant transformation in most somatic cells, neurons, which exhibit specialized mechanisms to protect cell fate, might react in a very different way. It is interesting to note that human adult tumours that arise from mature neurons seem to be extremely rare[119], and although cell cycle re-entry by neurons has been detected in post-mortem brain tissue from individuals with AD[120,121], a bona fide cell cycle completion has not been described. Thus, instead of initiating proliferation, the potential consequences of neuronal fate loss in neurons are senescence, metabolic stress and loosening of apoptotic brakes. Apoptosis competence is a feature of developmental neuronal precursors that is lost upon maturation but seems to be regained upon de-differentiation[32,122]. Once the apoptotic brake is lost, accumulated damage in the brain might function as the death stimulus. The accumulation of toxic amyloid-β species, which is often thought to be characteristic of AD but can also be found in healthy individuals[123], might act either as a trigger to induce DiD-RR or as a signal that initiates the cell death cascade in de-differentiated neurons, or both. Thus, one could conclude that AD can be described as a 'cancer of neurons', and it might thus be more useful to prevent age-related vulnerability to fate changes instead of trying to neutralize toxic species that are occasional and inevitable triggers that accumulate with age[124].

Partial fate loss and rejuvenation

Ageing is the biggest risk factor for many diseases (for example, neurodegenerative diseases and many types of cancer[6,125]), so an important question for research is whether ageing can be reversed to increase health span. In vitro reprogramming of somatic cells to a pluripotent state using the transcription factors OCT4, SOX2, KLF4 and MYC (OSKM) has already demonstrated that ageing is reversible to some degree[126] (Fig. 3). It was quickly noted that iPSCs generated from young and old human donors are largely indistinguishable[127–129], and this rejuvenation is preserved following differentiation of iPSCs into somatic cell fates[130,131]. The use of iPSCs to generate rejuvenated human organs, however, had severe drawbacks owing to potential tumorigenesis[132]. Interestingly, short-term OSKM expression in mice and in cells from older humans erases molecular ageing signatures from cells before they enter an iPSC fate[133]. Specifically, Ocampo and colleagues[133] showed that short-term OSKM induction in mouse fibroblasts ameliorated damage to the nuclear envelope, reduced markers of senescence, DNA damage and the stress response, and restored the epigenetic landscape for maintenance of heterochromatin. Short-term OSKM expression in aged animals improved several physiological functions of the pancreas, liver, spleen and blood,

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and also mitigated transcriptional, metabolic and epigenetic ageing signatures[134]. Partial rejuvenation was successfully used to restore vision in a mouse model of glaucoma[135] and it increased the regenerative capacity of adult mouse hearts after myocardial infarction[136]. The concept of OSKM rejuvenation, or chemical replacement equivalency, thus suggests that partial de-differentiation activates the regenerative potential of old and damaged cells by restoring a youthful gene expression[133.135,137,138]. This process is associated with a transient repression of somatic cell identity, which is very similar to damage-induced fate loss and is likely to hijack similar molecular machineries to activate cellular programmes of resilience and repair.

Such findings of rejuvenation and induction of regenerative capacities using partial de-differentiation raise new questions about the role of damage-induced de-differentiation in ageing and disease. Based on current literature, it remains unclear whether cell fate instability and partial fate loss are purely detrimental mechanisms that should be prevented or if they are necessary survival mechanisms. Rejuvenation of postmitotic neurons, in particular, might present further obstacles, as their cell fate is more tightly locked than that of other cells in the body[83]. Although this observation suggests that achieving rejuvenation of postmitotic neurons might be more challenging than rejuvenation of other somatic cell types, such rejuvenation of the brain is of great interest to prevent age-associated neurodegeneration. During development, neuronal precursor cells have the capacity to divide, differentiate to terminal neuronal fate, and undergo programmed cell death. Once mature, neurons must survive throughout the entire multidecade lifespan of a human being. They have thus established elaborate mechanisms to avoid cell death, the so-called apoptotic brakes[32,93,122]. The purpose of this unique resilience is to maintain circuit-based neuronal function. In theory, a partial de-differentiation to rejuvenate postmitotic neurons might result in the loss of synaptic connection established throughout life and lead to a regaining of apoptosis competence and neuronal cell death. In contrast, based on the fact that neurons are born during development and do not divide during their lifetime, ROS and other stressors might lead to an accumulation of damage to membranes and the nuclear pore with age. Proteins of the nuclear pore complex are exceptionally long-lived and rely on the cell cycle for renewal and repair; they are thus frequently disrupted in old postmitotic neurons[139]. Nuclear pore leakiness accumulates with age and is thought to contribute to neurodegeneration[131]. Theoretically, rejuvenation and initiation of the cell cycle could therefore renew nuclear pore structures, thus decreasing transcriptional noise caused by mis-located transcription factors. Based on the current literature, transcription factor-based rejuvenation can be viewed as a double-edged sword. De-differentiation of neurons in the context of ageing and pathology needs to be better understood to determine whether treatments should prevent or support partial fate loss. Targeted experiments inspired by new perspectives are required to address these questions.

Conclusions

Although different age-related diseases, such as cancer and neurodegeneration, have been thought to be based on very distinct disease mechanisms, we propose that age-related diseases might share more similarities than previously thought. Cell fate instability is a key feature of ageing and also seems to be a common denominator, and a fundamental cause,

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of cancer as well as neurodegeneration. Given that age-related neurodegenerative diseases are still incurable, the perspective devised here makes a case for further investigation into the interplay of different classes of disease mechanisms, including triggers, cell state shifts, ageing factors and cell fate loss, as this approach might help us to devise new strategies for therapeutic intervention.

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Glossary

Bivalent promoter chromatin marks

Activating and repressing histone modifications, such as histone 3 lysine 4 trimethylated (H3K4me3) and histone 3 lysine 27 trimethylated (H3K27me3).

Cellular senescence

An irreversible damage response to intrinsic or extrinsic stimuli in normal cells, which results in diverse phenotypic alterations including cell cycle arrest, an inflammatory secretome, metabolic alterations and epigenetic reprograming.

Epigenetic drift

Global stochastic deregulation of epigenetic patterns in response to environmental factors and ageing.

Health span

The number of years lived when a person is healthy and free from disease.

Metabolic drift

A change in preference for a set of specific metabolic pathways.

Nuclear pore

A protein-lined channel in the nuclear envelope that regulates the transport of molecules, such as transcription factors, into and out of the nucleus.

Proliferative arrest

The cessation of a cell proceeding through the cell cycle and dividing.

Transcriptional noise

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Variation in gene expression and transcriptional activity occurring among an otherwise homogeneous isogenic population of cells.

Warburg effect

A metabolic switch from oxidative metabolism to aerobic glycolysis, first described in cancer cells.

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Box 1

Clarification of terminology

Cell state

A cell state is a transient state of a cell in a dynamic system, where the cell can go into and out of the specific state. State shifts are often functional adaptations of cells. Examples of cells shifting between states include neuronal firing patterns during sleep and wakefulness[27], activation of glial cells[10,140], glucose-responding cells[12], temperature-sensing cells[11] and cells controlling circadian rhythm[13].

Cell fate

Cell fate describes developmental identity of a cell population, which does not allow dynamic shifting between different fates. A cell fate is, for example, a neuron, an astrocyte or a microglial cell. Within a cell fate there is high heterogeneity, as cells within the population can be present in different states. Thus, fate shifting is a permanent change of identity. Examples of fate shifting include carcinogenesis[141], trans-differentiation from astrocytes to neurons and the irreversible commitment to a differentiated cell type during development[142].

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Box 2

Cellular models used to assess neuronal state and fate changes

iPSC-derived neurons

The concept of fate loss in age-related neurodegenerative diseases is still incompletely understood, mostly owing to the inaccessibility of relevant cell types and adequate model systems to study the human brain. Starting in the early 2000s, neuroscience began to use human stem cells to model neurological diseases. Human induced pluripotent stem cells (iPSCs) generated from patient-derived fibroblasts have the ability to differentiate into all neural cell types and are thus a great model system for neurological diseases[143,144]. The great advantage of iPSCs is the expandability and high cell numbers obtained by iPSC neural differentiation, which allow drug testing and screenings[143–145]. Nevertheless, reprogramming to pluripotency resets the epigenetic state, resulting in epigenetically and functionally rejuvenated cells[127,146]. Thus, neuronal differentiation from iPSCs represents a great model system for the developing brain, but additional stressors are required to recapitulate age-associated disease phenotypes[147,148].

Directly reprogrammed neurons

By contrast, directly reprogrammed neurons have been shown to retain age-associated signatures and the epigenetic state of the donor, and therefore present a great model to study age-associated diseases[131,149,150]. Additionally, directly reprogrammed neurons recapitulate neurons of the adult human brain, as highlighted by the expression of 4R tau isoform[151], non-CG methylation[152] and transcriptomic correlation with adult stages of the human brain[92]. However, direct reprogramming skips developmental stages of neural development and directly changes cell fate from fibroblasts to neurons. Thus, directly reprogrammed neurons are not suitable for modelling changes in neurodevelopment[148]. In addition, owing to the slow proliferation of starting fibroblasts and the (necessary) lack of cell cycle during conversion, the yield of directly reprogrammed neurons is low and poses limitations for scalability of drug testing approaches.

Studying state and fate changes

In the context of state and fate changes, iPSC-derived neurons and directly reprogrammed neurons are complementary, and the model system chosen for experiments should be based on the pathology studied. For example, to establish physiological differentiation trajectories and deep phenotyping of developmental human cell fates, differentiating iPSCs seem to be the model system of choice. However, to address age-dependent state changes and damage-induced de-differentiation followed by cellular rejuvenation and re-differentiation, age-equivalent directly reprogrammed neurons can better capture the disease features, as the cells maintain the accumulated damage, contain age-specific metabolic dysfunctions and better resemble the adult human cell type[93,149]. Thus, directly reprogrammed neurons seem particularly useful to provide a more authentic reflection of the disease ‘in-a-dish’, and they might be informative in establishing predictive defined phenotypes (flagpost phenotype), which then can be assessed in more expandable models. In contrast, rejuvenated iPSC-derived disease

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models have the advantage of starting with a young healthy cell of a specific genetic makeup and, with the help of specific stressors, the defined flagpost phenotype can be induced. Thus, iPSC-derived and directly converted neurons are not necessarily exclusive but can be used as complementary systems. Both are based on a human-specific disease background and provide the opportunity to dynamically study the disease mechanisms, perform functional assays and assess response to drug treatments.

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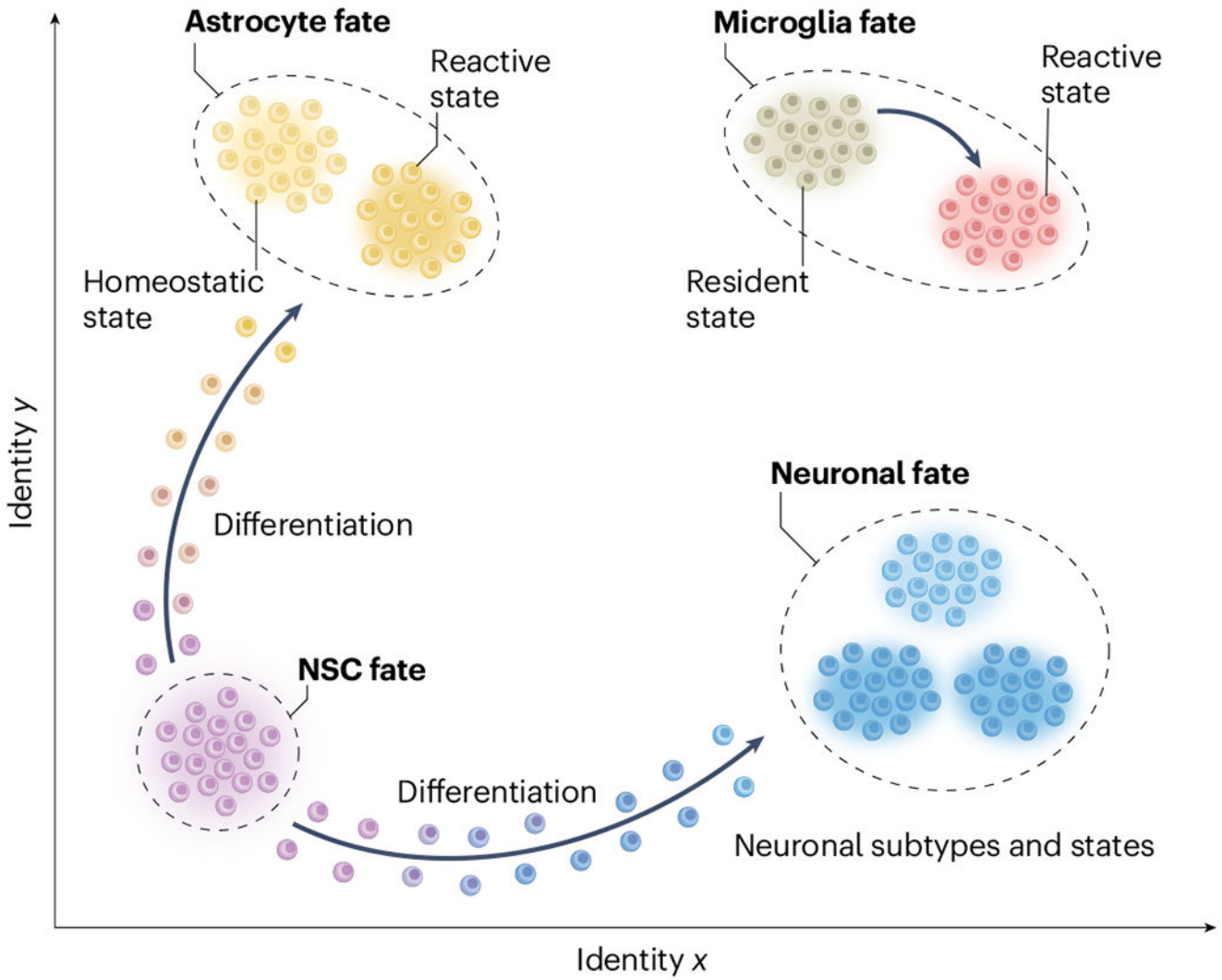

Fig. 1 |. Examples of cell states and fates in the brain.

Neural stem cells (NSCs) exist in different states and differentiate to specific cell fates – the neural or astrocyte fate – whereas microglial fate is derived from yolk sac-borne erythromyeloid precursors. Each fate forms a cloud of cell states, such as homeostatic and reactive astrocytes or microglia.

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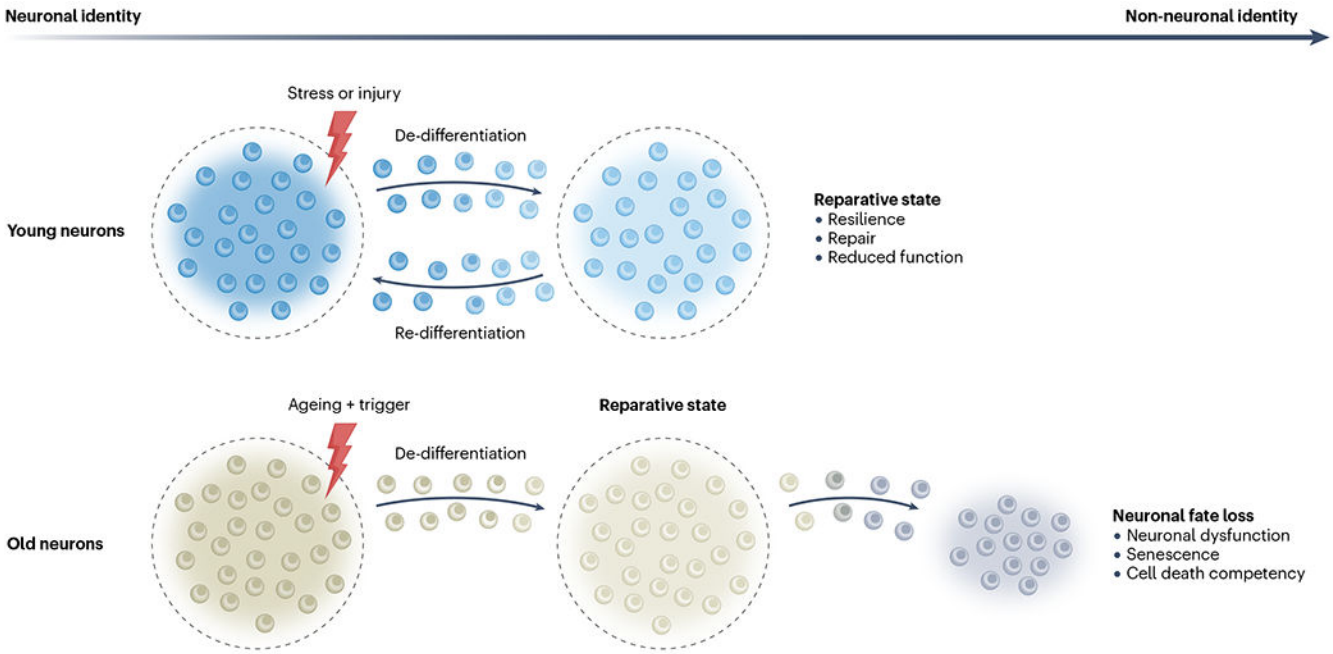

Fig. 2 |. Damage-induced state shifts in young neurons and fate loss in old neurons. Young neurons retain the capacity for damage-induced de-differentiation followed by cellular rejuvenation and re-differentiation, and respond with a reparative state to acute stress or injury. Old neurons present high levels of entropy and metabolic stress and therefore cannot exit the reparative state, resulting in neuronal fate loss and neurodegeneration.

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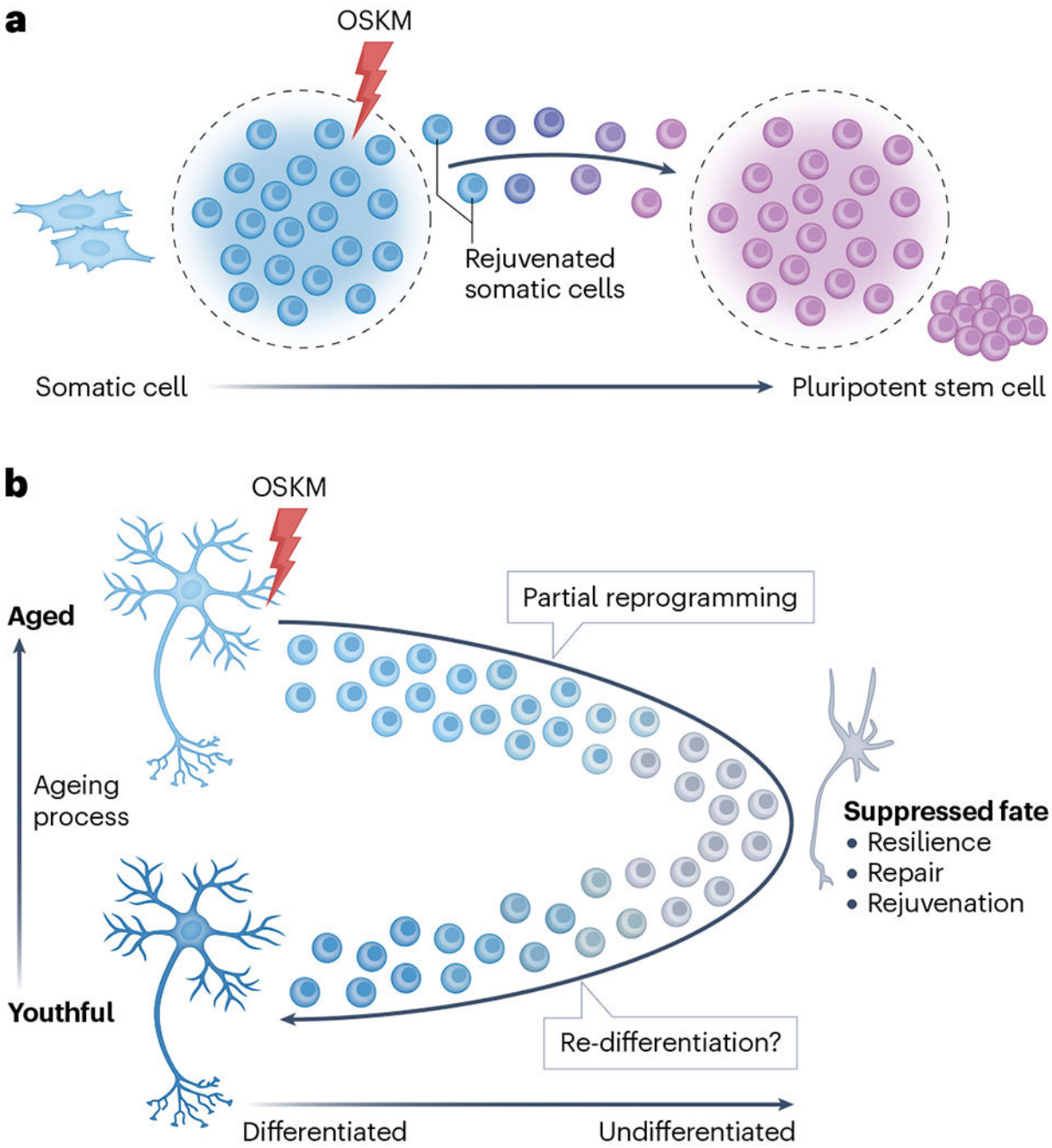

Fig. 3 |. Pluripotency reprogramming and partial reprogramming and rejuvenation for neurons. **a**, Reprogramming of somatic cells into pluripotent stem cells using the transcription factors OCT4, SOX2, KLF4 and MYC (OSKM). Somatic cells leave their identity cloud, resulting in a fate shift, losing their epigenetic information and being rejuvenated. **b**, Model of partial reprogramming of postmitotic neurons. Triggers, such as short-term OSKM exposure, result in partial reprogramming to a suppressed neuronal state of resilience, repair and

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rejuvenation. Eventual re-differentiation to mature neurons might result in repaired and fully functional neurons.